🥗 NutriAI – The Smartest AI Nutrition Assistant

Problem Statement #8 | IBM Hackathon Submission

IBM Granite
watsonx.ai

Problem Statement #8 – The Challenge

In an era where health awareness is growing, individuals increasingly seek personalised nutrition guidance. However, most existing tools provide **generic diet plans**, lack real-time adaptability, and fail to consider a person's holistic lifestyle, cultural preferences, allergies, and evolving health conditions.

Furthermore, dieticians and nutritionists face limitations in scaling personalised consultations due to time and resource constraints.

Generative AI presents a groundbreaking opportunity to revolutionize this space by enabling an **intelligent, interactive, and adaptive virtual nutrition assistant**. By leveraging NLP, multimodal understanding, and large-scale dietary databases, an AI-powered assistant can generate dynamic meal plans, recommend smart food swaps, and explain nutritional choices—all tailored to the individual.

Our Solution – NutriAI

NutriAI is a full-stack, IBM Granite-powered nutrition assistant that delivers personalised, adaptive, and culturally-aware dietary guidance. It bridges the gap between one-size-fits-all diet apps and in-person nutrition counselling.

Core Capabilities

- **Multimodal Input:** Text chat, voice input, and food image/label analysis via OCR
- **Personalised Meal Plans:** 1–14 day plans with macros, shopping list, and hydration tips
- **Smart Food Swaps:** AI-powered healthier alternatives with nutritional comparisons
- **Nutritional Explainer:** Deep science-backed explanations for any food or meal
- **Adaptive Learning:** Feedback-driven recommendations that improve over time
- **Health Profile:** BMI, TDEE calculation, allergy management, cultural preferences

Technology Stack

🤖 IBM Granite LLM

ibm/granite-3-1-8b-base via watsonx.ai SDK
IBM Cloud Lite (free) tier

Backend

Python 3.14, Flask 3.0, Flask-CORS
Gunicorn (production WSGI)

🌐 Frontend
Vanilla HTML5/CSS3/JavaScript
IBM Design System inspired UI

👁 Computer Vision
Pillow image processing
Tesseract OCR for label reading

🗄 Food Database
Built-in DB (30+ foods)
USDA FoodData Central API

☁ IBM Cloud
watsonx.ai project
IAM API key authentication

System Architecture

Layer	Component	Description
①	Presentation	Single-page web app with 8 interactive panels
②	API Gateway	Flask REST API – 13 endpoints
③	Agent Core	Intent Router, Profile Manager, Prompt Builder, Memory Store
④	IBM Granite LLM	granite-3-1-8b-base via watsonx.ai (temp:0.7, max_tokens:2048)
⑤	Data & Tools	Food DB, USDA API, Nutritionix API, Tesseract OCR
⑥	Infrastructure	Python venv, Gunicorn, IBM Cloud Lite, .env secrets

IBM Granite Integration

NutriAI uses the **ibm-watsonx-ai SDK** to connect to IBM Granite models hosted on IBM Cloud watsonx.ai. Six specialised prompt templates are used — each injecting the user's complete health profile as context to ensure every response is personalised.

> **Mandatory IBM Technology Used:** IBM Cloud Lite services + IBM Granite (ibm/granite-3-1-8b-base) via watsonx.ai SDK. All AI responses are generated by IBM Granite with adaptive prompt engineering.

Key API Endpoints

Endpoint	Method	Feature
/api/chat	POST	Conversational nutrition Q&A
/api/meal-plan	POST	Personalised meal plan generation
/api/food-swap	POST	Smart food swap suggestions

/api/explain	POST	Nutritional explanation for any food
/api/analyse-image	POST	Food photo / grocery label OCR analysis
/api/suggestions	GET	Adaptive personalised meal suggestions
/api/feedback	POST	Star-rating feedback for adaptive learning
/api/food-lookup	GET	Instant nutritional data lookup

How to Run

1. Clone the repository from GitHub
2. Create virtual environment: `python -m venv venv` and activate it
3. Install dependencies: `pip install -r requirements.txt`
4. Copy `env.example` to `.env` and fill IBM credentials
5. Run: `python app.py`
6. Open: `http://localhost:5000`